

Foundations of Data Analytics and Data Science

i) What is a database

* It is an organised collection of information stored on a computer so it can be easily accessed, managed and updated

Why database are essential in the modern world

* because they help people and organisations store and find information quickly and accurately

Two real-world examples of where data are used

- a) Banks use database to store customer accounts and transactions
- b) Schools use database to store learners marks, attendance and personal information

2) What is a data warehouse

* is a large storage system that collects and stores data from different sources so it can be analysed and used for decision making

ii) How does it differs from a regular database

* a regular database is mainly used to store and manage current information while a data warehouse is designed to store large amounts of historical data for analysis

iii) Why would companies use a data warehouse

* Study trends, create reports, compare performance and make better business decisions

3, What is a data lake
* It is a large storage system that keeps different types of data, such as text, images, videos and numbers in their original form

ii A data lake stores raw and unprocessed data, while a data warehouse stores organised and processed data that is ready for analysis

iii A business would prefer a data lake when it needs to store large amounts of different types of data and wants flexibility to analyse the data later, for example, a company can use a data lake to store social media posts, customer information, videos and sales data.

4, What is a data Mart

* is a smaller storage system that contains data for specific department or business area

ii It is usually created from a data warehouse and focuses on a particular area such as sales, finance or marketing

iii Data marts are typically used by department managers, business analyst and employees who need quick access to information for their specific area

5, Structured data is organised in a fixed format such as rows and columns
e.g a school spreadsheet containing learners names and marks

ii Semi structured data does not follow a strict table format but has some organisation such as tags or labels
e.g. an email or JSON file

iii Unstructured data has no fixed or organised format
e.g. a photo, video, or a social media post

6. What is data modeling

* is a process of planning how data will be organised, stored and connected in a system

ii It is important because it helps make data organised, accurate, easy to access and less likely to be duplicated

iii Two types of data models are

1. Relational model → organises data into tables made up of rows and columns

e.g. a school database with separate tables for learners, subjects and marks

2. Hierarchical model → organises data in a tree-like structure where information is arranged in parent and child relationships

Ti What is data mining

* is the process of finding useful patterns and information in large amounts of data

✓ MongoDB - a NoSQL database that stores data in flexible document formats rather than traditional tables

eg applications that handle changing or large amounts of data such as social media platforms

9. What is ETL

i. Extract - data is collected from different sources such as databases, spreadsheets or websites

ii. Transform - the data is cleaned, corrected and organised into a useful format

iii. Load - the prepared data is stored in a database or data warehouse where it can be analysed

iv. Why ETL is important

* ETL helps business turn raw data into clean organised and useful information making it easier to analyse data and make better decisions

10. Common Data formats used in Analytics

1. CSV → comma-separated values its a simple text format that stores data in rows and columns, it is useful for small datasets and sharing data between application

2. JSON → JavaScript Object Notation it stores data in a structured format using key value pairs it is used for web application and APIs

12) Machine Learning ML

is a branch of AI that allows computers to learn from data and improve their performance without being explicitly programmed for every task

ii Difference of traditional programming
~~the~~ traditional programming uses rules written by a programmer to produce an output, machine learning uses data to learn patterns and make predictions or decisions

iii Three main types of Machine Learning

1. Supervised Learning

learns from labelled data where the correct answers are provided

e.g predicting house prices

2. Unsupervised Learning

finds patterns or groups in data without labelled answers

e.g grouping customers based on their buying behaviour

3 Reinforcement Learning

learns by receiving rewards or penalties for actions and improves through trial and error

e.g training a robot to navigate a room

13) Deep learning

is a type of machine learning that uses neural networks with many layers to learn patterns from large amounts of data

3 Parquet → a column based format designed for efficient storage and fast data processing it is best for large datasets and big data analytics

4 Avro - a compact row based format that stores data together with its schema, it is used for data pipelines and systems that exchange large amounts of data

ii) What is Artificial Intelligence AI

- is the ability of computers or machines to perform tasks that normally require human intelligence such as learning, reasoning and problem solving

ii) Narrow AI (Weak AI)

- designed to perform a specific task or a limited range of task, it cannot think beyond what it was designed to do

e.g Siri which can answer questions and perform certain commands

iii) General AI (Strong AI)

- it is a theoretical type of AI that would be able to understand, learn and perform many different tasks at a human like level

e.g a hypothetical machine that could learn any subject, solve different problems and perform any intellectual task like a human

Relationship

1. Artificial Intelligence AI is the broad broad field of making machines perform tasks that require human intelligence
2. Machine Learning ML is a part of AI where computers learn from data
3. Deep Learning is a part of ML that uses multi-layered neural networks.

4. Neural networks

neural networks are computer systems inspired by the human brain, they are made up of connected nodes neurons arranged in layers, they take in data, process it through the layers, identify patterns and produce an output, during training the network adjusts its connections to improve its accuracy

14. Natural Language Processing NLP

is a branch of AI that enables computers to understand, interpret and respond to human language

Three real-world application of NLP

1. Voice assistants - Siri and Google Assistant understand spoken commands and respond to users
2. Chatbots - customer service chatbots understand questions and provide answers

3 Autocorrect and predictive text

- smartphones predict words and correct spelling while you type

15; LLM → A Large Language Model is an AI system trained on large amounts of text to understand and generate human language

ii Difference

→ an LLM can perform many languages tasks, while a standard ML model is usually designed for a specific task

iii Examples

- GPT - used for answering questions and generating text
- Gemini - used for answering questions and creating content

16; Generative AI is a type of AI that creates new content such as text, images, audios or videos, it works by learning patterns from large amounts of data and using them to generate new content based on a users instructions

Three tools

1. ChatGPT → generates and explains text
2. Dall-E → creates images from text description
3. Google Gemini → answers questions and generates text

17; A training database is a collection of data used to teach an AI model how to recognize patterns and make prediction

ii Good quality data is important because it helps the AI give accurate and reliable results, if the data is biased or incorrect, the AI may learn wrong patterns, make unfair decisions or give inaccurate answer

18; Computer vision is a type of AI that allows computers to understand and interpret images and videos and it analyzes pixels to identify objects, faces or patterns

ii Use Cases

1. facial recognition on smartphones

2. Self-driving cars detecting roads and obstacles

20
19; An API key is a unique code used to access an API and identify or authorize a user or application

ii Purpose → controls and authenticates API access

iii Protection → store is securely not in public code or repositories

iv If exposed others may misuse the API, causing unauthorized access or unexpected cost

19 An API allows different software system to communicate and share data

ii Analogy → an API is like a waiter who takes your order to the kitchen and brings back your food

iii Importance → APIs helps modern software and data systems connect, exchange information and work together efficiently

2. REST API is an API that allows applications to communicate over the internet

i. REST stands for Representational State Transfer

ii GET - Retrieves data

iii POST - Creates new data

iv PUT - Updates existing data

v DELETE - Removes data

2. JSON (JavaScript Object Notation) is a simple format for storing and exchange data, it is commonly used with APIs because it is easy for humans and computers to read

Example

"name": "Peter"

"age": 15,

"city": "Johannesburg"

23; Prompt Engineering
It is a skill of writing clear and specific instructions for AI tools like ChatGPT or Claude to get better and more accurate results

!! Example 1

→ Poor "Tell me about AI"

→ Improved. "Explain AI in simple words and give 3 real world examples"

!!! Example 2.

→ Poor "Write a report"

→ Improved "Write a short 200 word report about the benefits of AI in education"

24; What is a MAP function

→ a map function applies the same operation to each item in a list or collection and it returns the results

!! Python example

```
numbers = [1, 2, 3]
```

```
result = list(map(lambda x: x*2, numbers))  
print(result)
```

Output [2, 4, 6]

25, Kthot is cloud computing

- cloud computing is the use of internet based servers to store data, run applications and provide computing services

ii) IaaS (Infrastructure as a Service): provides virtual servers, storage and networks
e.g Amazon EC2

iii) PaaS (Platform as a Service): provides a platform for developing and running applications
e.g google App Engine

iv) SaaS (Software as a Service): provides ready to use software over the internet
e.g Microsoft 365

26, Kthot is a version control system

- A version control system (VCS) tracks changes to file and code, allowing developers to work together and restore previous versions

i) Git is a popular version control system widely used by developers and data analysts

iii) Commit: Saves a set of changes to the project history

iv) Branch: A separate line of development used to work on changes without affecting the main project

27 Kthot is a Jupyter Notebook

- is an interactive tool used to write and run code,

view results and document analysis in one place

"It is most commonly used with Python, it is popular among Data Analyst and Data Scientist because it makes it easy to analyze data, create visualizations, run code step by step and explain results

28 What is Python

- Python is a popular easy to learn programming language widely used in data analytics, data science, automation and machine learning

Three common Python libraries for data work are:

1. Pandas - used for cleaning, organizing and analyzing data

2. NumPy - used for numerical calculations and working with arrays

3. Matplotlib - used for creating graphs and data visualizations

29 Data privacy is protecting people's personal information. It is important because it prevents misuse of data. GDPR is a European law that protects personal data and privacy

30 Data Ethics means using data fairly and responsibly

Examples of unethical use:

1. Sharing personal information without permission
2. Using biased AI that treats people unfairly
3. Creating fake information using AI

31 Data governance, is managing data properly, its key components include

- data quality
- data security
- data privacy
- data policies
- roles and responsibilities

32 Bias in AI occurs when an AI system produces unfair results

e.g. A hiring AI may unfairly reject women because it was trained using mostly men's application

33 Differences Between Data Roles

Role	Main Job	Tools
Data Analyst	Studies data and find trends	Excel, SQL
Data Scientist	Builds predictions using data	Python, ML
Data Engineer	Collects and manages data	SQL, ETL
AI Engineer	Develops AI systems	Python AI tools

34 Business Intelligence uses data to help business make decisions

9 Power BI → creates reports and dashboards

Tableau → creates charts and visualization

51 found AI and machine learning interesting because computers can learn from data and make predictions, I would like to learn more about how AI works